

24M15CA111 – Data Structure Using C Lab
ODD 2025
Week 6
Practice Lab
[CO2/CO3]

Instructions:

1. All students must save all their programs with the nomenclature (Enroll No_W6_QuestionNo.cpp). Also, store the Outputs as well.
2. Upload them as per the instructions given by your lab faculty.

Topics: Structure/ Union/ Pointer

Lab Questions:

Lab 11: Structure / Pointer

1. Write a program using structures to read and display the information about a student.
2. Write a program to read and display the information of a student using a nested structure.
3. Write a program to read and display the information of all the students in a class. Then edit the details of the ith student and redisplay the entire information
4. Write a program to read, display, add, and subtract two distances. Distance must be defined using kms and meters.
5. Write a program to initialize the members of a structure by using a pointer to the structure.
6. Define a structure to store the name, an array marks[] which stores the marks of three different subjects, and a character grade. Write a program to display the details of the student whose name is entered by the user. Use the structure definition of the first question to make an array of students. Display the name of the students who have secured less than 40% of the aggregate.
7. Declare a structure fraction that has two fields—numerator and denominator. Create two variables and compare them using function. Return 0 if the two fractions are equal, -1 if the first fraction is less than the second and 1 otherwise. You may convert a fraction into a floating point number for your convenience.

Lab12: Union ,Pointer

1. Write a program using structures to read and display the information about a student.
2. Write a program to calculate the area of one of the geometric figures—circle, rectangle or a triangle. Write a function to calculate the area. The function must receive one parameter which is a structure that contains the type of figure and the size of the components needed to calculate the area must be a part of a union. Note that a circle requires just one component, rectangle requires two components and a triangle requires the size of three components to calculate the area.
3. Design a union that can store either a reporter's name (string) or their ID (integer). Write a program that uses a **function** to display whichever detail is stored, based on user choice. This example relates to a newsroom management system, where either a reporter's name or an ID is required for quick searches. By using unions, memory is saved when only one type of data is necessary. This problem demonstrates the real-life application of unions in media systems that need efficient storage while handling large reporter databases.
4. Create a union that can store money in different formats: rupees (integer), dollars (float), or euros (float). Write a program that uses functions with union pointers to read values and perform conversions from one currency to another. This problem has a real-world connection with foreign exchange counters or banking systems where currency conversion is a daily task. It demonstrates how unions can save memory while handling multiple representations of the same data. It also allows practicing pointer usage with functions, making it a practical learning exercise.
5. Define a union that can store either mobile model number (integer) or laptop RAM (integer). Create an array of devices, store details for at least five devices, and display the data using pointer traversal. This example is taken from electronics inventory management, where either mobile or laptop details are stored in the same database. Since not all fields are required simultaneously, using unions saves memory. The use of pointers and arrays makes the problem an effective practice in data organization and retrieval.